

Pointers

Lab questions, set 5

41. Write a C program to store a string using:

- i. A character array
- ii. A character pointer

Print the string in both cases and analyze the difference in memory allocation used by each declaration.

Assume data:

"ComputerScience"

Also mention why the indirection operator (*) is not required while printing the string.

42. Write a C program to determine the length of a string by traversing it using a pointer variable only.

Do not use any string handling library functions.

Assume data:

char str[] = "National Institute of Technology";

Explain how the pointer moves until the null character ('\0') is encountered.

43. Write a C program to store multiple city names using an array of character pointers and display them.

Assume data:

{"Delhi", "Mumbai", "Chennai", "Kolkata"}

Additionally:

- i. Calculate the total memory used by this approach
- ii. Compare it with a 2D character array approach

44. Write a C program to clearly demonstrate the difference between:

- i. int *p[3]
- ii. int (*p)[3]

Assume data:

int a = 5, b = 10, c = 15;

int arr[3] = {2, 4, 6};

Explain:

- a) What each declaration represents
- b) How operator precedence affects pointer interpretation

45. Write a C program that performs addition, subtraction, and multiplication of two numbers using a single function that returns results through pointer parameters.

Assume data:

`int a = 12, b = 4;`

Explain why pointers are required instead of return statements.

46. Write a C program to demonstrate call by reference by passing the address of a variable to a function and modifying its value.

Assume data:

`int salary = 125000;`

Increase salary by 10% inside the function and explain:

- i. Why the change reflects in the calling function
- ii. Difference between call by value and call by reference

47. Write a C program to sort an integer array in ascending order using pointer arithmetic.

Array indexing ([]) should not be used inside the sorting logic.

Assume data:

`int arr[] = {9, 4, 7, 1, 6};`

Explain how pointer arithmetic replaces array indexing.

48. Write a C program in which a function returns a pointer to the larger of two integers.

Assume data:

`int x = 40, y = 75;`

Additionally:

- i. Explain why returning the address of a local variable is unsafe
- ii. Justify why returning x or y is valid

49. Write a C program that uses a function pointer to perform arithmetic operations based on user choice.

Assume data:

Operands: 10 and 5

Operations: Addition, Subtraction

50. Void Pointer and Type Casting in Memory Interpretation

Write a C program to demonstrate the use of a void* pointer to store addresses of different data types and access them using type casting.

Assume data:

`int a = 100;`

`float b = 25.5;`